Unscrambling traffic congestion and increasing sustainability in special urban intersection

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Abstract: Traffic congestion in India's densely populated urban centres is a major challenge exacerbated by rapid urbanization and a surge in vehicles surpassing infrastructure growth. Gridlocks and slow-moving vehicles are common due to inadequate road networks and poorly planned intersections, leading to longer commutes, productivity loss, and heightened pollution and safety concerns. Mitigation efforts include infrastructure upgrades and sustainable transportation promotion, implementing Indian Road Congress (IRC) signal design guidelines for intersection optimization and traffic flow management. Pedestrian safety remains critical, requiring comprehensive strategies integrating public transportation enhancement, road infrastructure optimization, and alternative travel modes promotion. Utilizing advanced technology for traffic monitoring, alongside urban planning initiatives prioritizing efficient land use and mixed-use developments, is crucial. Furthermore, policies promoting telecommuting and flexible work schedules can help alleviate the burden of peak-hour traffic. Flyovers offer relief by allowing vehicles to bypass congested areas, improving traffic flow. Congestion indirectly contributes to pollution, emphasizing the importance of catalytic converters in reducing harmful emissions from vehicle exhaust. This comprehensive approach aims to create smoother and more sustainable transportation systems while addressing congestion and pollution issues effectively.

Keywords: Traffic volume, pollutions, IRC , PCU Values

1 Introduction

Traffic congestion in India presents a significant challenge, especially in densely populated urban areas. Rapid urbanization and economic growth have led to a surge in the number of vehicles on Indian roads [2] , surpassing the expansion of infrastructure. The result is chaotic traffic scenarios marked by gridlocks, slow-moving vehicles, and frayed nerves [3]. In many cities, inadequate road networks, poorly planned intersections, and a lack of proper traffic management exacerbate the problem. Moreover, the proliferation of motor vehicles, including cars, motorcycles, and auto-rickshaws, coupled with limited public transportation options, further compounds the issue. This congestion not only leads to longer commute times and decreased productivity but also contributes to heightened air pollution levels and

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road safety concerns [8]. Efforts to alleviate congestion include infrastructure upgrades, the introduction of mass transit systems, and initiatives promoting sustainable transportation alternatives. However, addressing traffic congestion in India demands comprehensive urban planning strategies, effective enforcement of traffic regulations, and investments in public transportation infrastructure to ensure smoother and more efficient mobility for its citizens. Pedestrian cases in traffic congestion highlight the immersed risks pedestrians face in crowded urban environments [9]. With limited space and chaotic road conditions, pedestrians navigating through congested areas are vulnerable to accidents. Factors such as distracted driving, poor visibility, and hurried crossing behaviours contribute to the prevalence of pedestrian incidents amidst traffic congestion [7].

Signal design involves the planning and configuration of traffic signals at intersections to facilitate the safe and efficient movement of vehicles [1], pedestrians, and cyclists while minimizing congestion and improving overall traffic flow. Signal design controls congestion by optimizing signal timing and coordination to regulate traffic flow efficiently. Through data analysis and modelling, signal design identifies congestion patterns and implements strategies such as adaptive signal control systems to dynamically adjust signal timings based on real-time traffic conditions [4]. Additionally, pedestrian and cyclist accommodations, along with synchronized signal networks, contribute to smoother traffic flow and reduced congestion at intersections.

Traffic congestion significantly impacts pollution levels through increased vehicle emissions and inefficient traffic flow [8], worsening air quality and environmental conditions in urban settings. Efficient traffic management strategies, such as optimized signal timings and enhanced public transportation [7], help alleviate congestion, reduce idling time, and ultimately lower pollution levels. Additionally, investments in sustainable transportation infrastructure, like bike lanes and pedestrian-friendly pathways, offer alternative modes of travel, reducing dependence on private vehicles and further mitigating pollution associated with traffic congestion. While catalytic converters play a role in reducing emissions and improving air quality, addressing traffic congestion requires comprehensive transportation management solutions and infrastructure enhancements.

2 Objective

The objective is to optimize traffic flow, alleviate congestion, and promote environmental sustainability through the implementation of innovative intersection designs, traffic management strategies, and sustainable transportation solutions.

3 Problem description

3.1 Issue due to Congestion

Traffic congestion is a multifaceted issue arising from various factors. Firstly, the sheer volume of vehicles, particularly during peak hours or in densely populated areas, overwhelms roads, causing congestion. Limited road capacity exacerbates this problem, leading to bottlenecks and traffic snarls, especially where infrastructure is lacking [11]. Additionally, incidents like accidents and breakdowns disrupt traffic flow, compounded by inadequate alternative routes. Poor road design, with insufficient lanes or confusing signage, further contributes to congestion, as does inefficient traffic management, including poorly timed signals. Urbanization and population growth fuel increased vehicle ownership, while dependency on single-occupancy vehicles worsens congestion. Inadequate public transportation options also push more people towards private vehicles.

Behavioural factors such as aggressive driving and distracted behaviour compound congestion by disrupting traffic flow [5]. Altogether, congestion negatively impacts productivity, air quality, and quality of life in urban areas, underscoring the urgent need for effective transportation planning, infrastructure upgrades, and alternative transportation solutions to alleviate this pervasive issue.

3.2 Sustainability in traffic congestion involves

Sustainability in tackling traffic congestion entails a comprehensive approach aimed at minimizing environmental impact, optimizing transportation efficiency, and enhancing overall quality of life [15]. This involves a concerted effort to promote alternatives to single-occupancy vehicles such as public transit, biking, walking, and carpooling. It also entails improving infrastructure by developing dedicated lanes for buses, bikes, and pedestrians to ease congestion. Leveraging technology for smart traffic management, including real-time monitoring and dynamic routing, can further enhance traffic flow. Integrating land use planning with transportation initiatives helps minimize lengthy commutes and supports sustainable modes of travel. Addressing environmental concerns such as air pollution by promoting cleaner transportation modes is also crucial. Encouraging behavioural change through public education and incentives for eco-friendly commuting habits reinforces sustainable transportation practices [14]. Additionally, policy measures like congestion pricing, fuel taxes, and emission standards play a vital role in managing demand and reducing congestion sustainably, while considering the long-term environmental and societal implications. By adopting these strategies and policies, communities can work towards alleviating traffic congestion in a sustainable manner, fostering healthier and more livable urban environments [12].

3.3 Pollution in traffic congestion involves

Traffic congestion not only results in frustrating delays but also presents significant environmental challenges. In stop-and-go traffic, vehicles operate less efficiently, leading to increased emissions of pollutants such as carbon monoxide, nitrogen oxides, and particulate matter [6]. Idling engines worsen the issue by releasing pollutants into the air without moving effectively. Consequently, urban areas experience worsened air quality, which in turn can result in health issues such as respiratory problems and cardiovascular diseases. Additionally, congestion drives up fuel consumption as vehicles spend more time on the road, burning more fuel and emitting more pollutants. This environmental impact extends beyond immediate health concerns, contributing to broader ecological consequences and exacerbating climate change [12]. Addressing traffic congestion through improved transportation infrastructure and the promotion of alternative modes of transportation is crucial for mitigating pollution and safeguarding overall air quality in urban environments.

4 Methodology

4.1 Selection of junction

The selection of Puthupally, Kottayam, as a focal point for tackling traffic congestion is likely influenced by several key factors. Firstly, the area's population density could contribute to heavy vehicular traffic, either due to urban expansion or its strategic location,

resulting in congestion problems (6). Additionally, if Puthupally encompasses commercial centers, educational institutions, or residential zones, the influx of traffic during peak hours could exacerbate congestion issues. Moreover, Puthupally might function as a transportation hub, with major roads or highways intersecting, leading to congestion as vehicles converge from various directions. Concerns regarding traffic accidents or bottlenecks may also prompt efforts to address congestion in Puthupally, prioritizing safety and efficient traffic management. Furthermore, community feedback from local residents, authorities, or businesses expressing concerns about congestion could drive initiatives to alleviate the issue. In summary, the decision to focus on Puthupally for congestion alleviation likely arises from its significance as a transportation node, high traffic volumes, and the community's demand for smoother traffic flow and enhanced urban mobility.

4.2 Field survey and Data collections

Puthupally, located in the Kottayam district of Kerala, India, stands out as one of the busiest urban intersection areas. It serves as a convergence point for four main roads. They are Kottayam, Manarcadu, Njaliyakuyil, and Karukachal. Puthupally is one of the famous special urban intersection areas. Because high court is passed by puthupally route and near by a famous historical church placed them after then this place is specialised. Bus stations and taxi stands are situated in close proximity to the junction, exacerbating congestion. Additionally, inadequate road width, lack of traffic signals, insufficient parking space, unauthorized roadside trading, and a dearth of pedestrian amenities further compound the traffic challenges in the area. These factors, unique to Puthupally, contribute significantly to traffic issues, particularly during peak hours. To delve deeper into these concerns, We conducted a direct field survey to collect pertinent data. This included compiling accident records from 2021 to 2023 and conducting peak-hour volume counts of vehicles in all directions to measure traffic flow. To streamline data analysis amid mixed traffic conditions, we utilized a standardized unit known as the Passenger Car Unit (PCU) for all vehicle counts. We closely observed vehicle turning movements in 12 directions during morning and evening peak hours. Furthermore, we documented pedestrian movement patterns and tallied both pedestrians and vehicles using a basic hand counter.

4.3 Data Analysis and Synthesis

To understand the reasons behind traffic issues at the Puthupally junction. We've analyzed the collected field data and used deductive reasoning. The results of this analysis are outlined in the following sections.

4.3.1 Accident Analysis

The analysis of accident data collected over the three-year period reveals the number of victims who died, were injured, and affected by grief during the accidents, as illustrated in Figure 4.1. It's clear from the figure that there has been a notable increase in the number of victims from 2020 to 2023, with the exception of the grief category. Fatal accidents were found to be relatively low in number, while incidents resulting in significant personal losses were notably high in the grief category. Additionally, a detailed examination of the accident data revealed that pedestrians were disproportionately affected by the accidents.

4.3.2 Turning movement survey

The analysis of vehicle turning movements at the Ettumanoor intersections reveals key findings presented in Fig. 4.2. Notably, the turning movement at Ettumanoor-Kottayam experiences heavy traffic volume. Similarly, routes to and from Ernakulam and Pala also face high traffic density. The traffic jam is caused by missing road markings, enough medians, and proper traffic signs and signals.

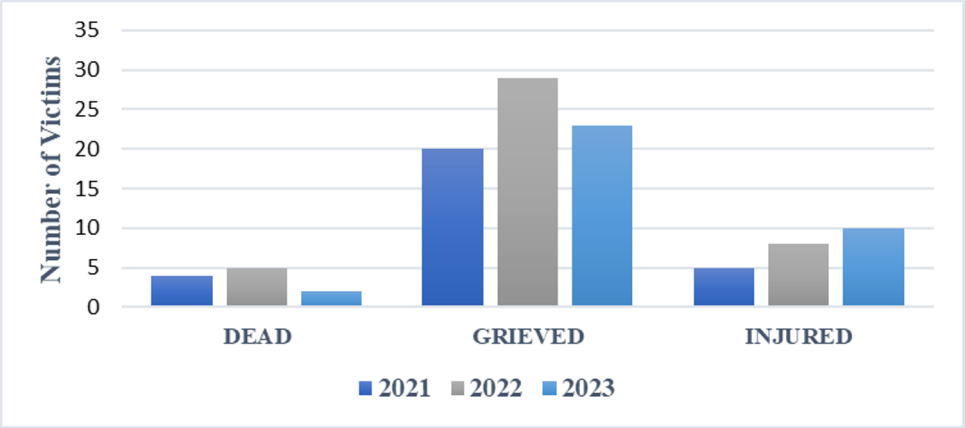

Fig. 4.1. Result of accident analysis.

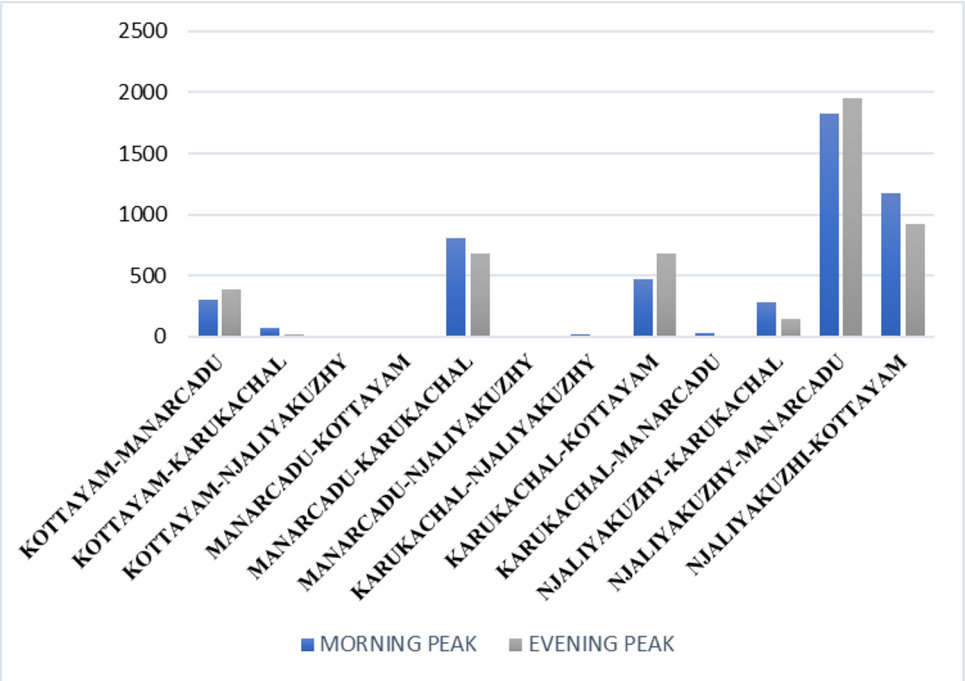

Fig. 4.2. Turning movement at different routes during the peak hours.

4.3.3 Pedestrian Survey

Pedestrians, as the most vulnerable road users, this rise in victims is alarming, were the focus of our survey, with an emphasis on analyzing their movements across and along

roads. Our findings revealed that the average pedestrian speed ranged from 1 to 1.5 meters per second. Figures 4.3 and 4.4 Illustrate the hourly pedestrian movements towards four routes during morning and evening peak hours. Both graphs show more pedestrians using the Manarcadu route. In particular, the hourly volume of pedestrian movement surpasses the permissible limit of 500 during peak hours. Furthermore, pedestrian volumes exhibit minimal variation across both cross and lateral directions. Field observations indicate the absence of pedestrian crossings or footpaths along the Manarcadu route.

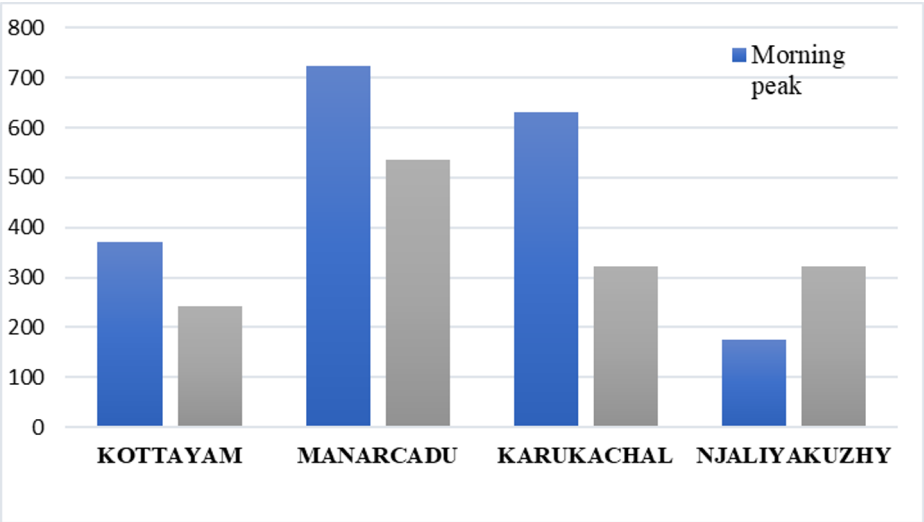

Fig. 4.3. Cross movement of pedestrians.

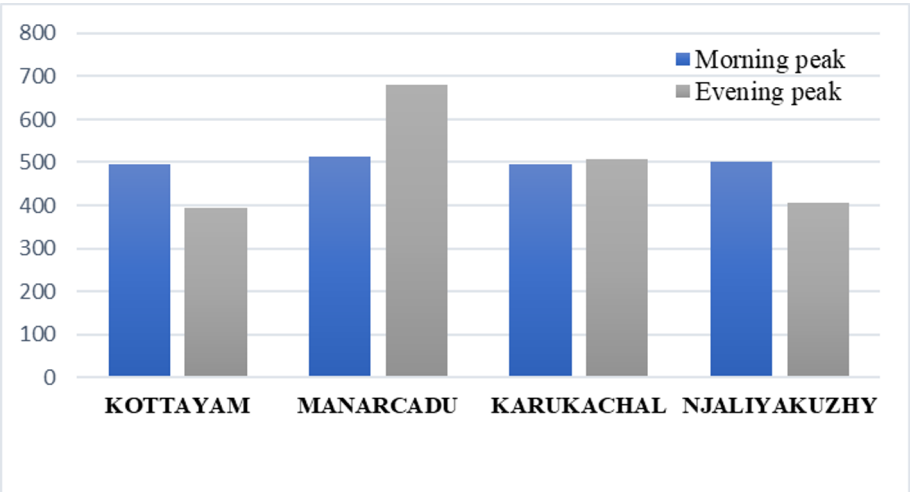

Fig. 4.4. Lateral movement of pedestrians

5 Proposed Modifications and Improvements

After analyzing the data and identifying key problems, we propose several methods to improve the Puthupally urban intersection. These modifications for different routes are outlined in Table 1. Additionally, we recommend constructing footpaths with a width of 1.5

meters, ensuring proper drainage, and installing medians with a width of 1 meter along all routesAdditionally, we propose installing adequate road markings, traffic signs, and signals. Traffic signals were designed for the two junctions of the Puthupally intersection based on collected traffic volume and standard procedures, with details provided in Table 2. Furthermore, we suggest installing 'no parking' boards at relevant spots and enforcing strict restrictions on unauthorized parking by imposing heavy penalties. It is also recommended to impose strict regulations on hawking and trading along roadside areas, with proper measures for the rehabilitation of roadside traders.

Table 1. Proposed modification along different routes at the intersection

Sl	Route	Proposed Modification
1	Manarkadu	Channelizing island, Turning radii improvement, pedestrian foot path and traffic signal
2	Kottayam	Channelizing island, Turning radii improvement, pedestrian foot path and traffic signal
3	Njaliyakuzhiyil	Channelizing island, pedestrian foot path and traffic signal
4	Karukachal	Channelizing island, pedestrian foot path and traffic signal

Table 2. Proposed signal characteristics

Sl	Phase	Cycle length	Green time	Amber time	Red time	Pedestrian crossing time
1	Phase 1 (Manarcadu to Kottayam & Manarcadu to Karukachal)	87	23	4	14	4
2	Phase 2 (Kottayam to Karukachal & Kottayam to Manarcadu)	87	12	4	14	4
3	Phase 3 (Njaliyakuzhy to Kottayan, Njaliyakuzhy to Karukachal & Njaliyakuzhy to Manarcadu)	87	28	4	14	4

6 Suggestion and Recommendations

6.1 Flyovers and Overpasses.

Puthuppally Junction experiences heavy traffic congestion due to the convergence of multiple roads or high volumes of traffic from nearby areas. A flyover would alleviate this issue by facilitating smoother traffic flow, enabling vehicles to bypass the junction without being subjected to traffic signals or navigating through crowded intersections.

- Safety Concerns: The high traffic volume at the junction poses significant safety hazards, including accidents and pedestrian-vehicle conflicts. Implementing a flyover offers a safer route for vehicles to traverse, reducing the risk of collisions with other vehicles or pedestrians.
- Enhancing Connectivity: Puthuppally Junction can be a vital link connecting various parts of the city or region. Establishing a flyover could enhance connectivity and facilitate smoother movement of vehicles, thereby improving overall transportation efficiency

6.2 Footpaths and Sidewalks

Establishment of footpaths in Puthuppally Junction is crucial for creating a safe, accessible, and vibrant urban environment that promotes pedestrian mobility, public health, and community well-being.

6.3 Underpasses and Tunnels

Establishing an underpass in Puthuppally Junction can significantly improve pedestrian safety, traffic flow, accessibility, and urban development, while also enhancing the overall quality of the urban environment

6.4 Bypass Roads

- Puthupally Junction may experience heavy traffic due to its location on major routes or because it serves as a convergence point for multiple roads. A bypass helps divert through-traffic away from the junction, reducing congestion and improving local traffic movement.
- Safety Improvement: High traffic congestion can lead to increased accidents and road hazards. By redirecting through-traffic away from the junction, a bypass can contribute to enhancing road safety for both motorists and pedestrians at the junction.

7 Conclusion

The rapid population growth and urbanization in India have triggered a significant surge in vehicle usage, culminating in pervasive traffic congestion issues. To really tackle this problem, a detailed study focused on a super important spot where roads meet in Puthupally. Field surveys thoroughly collected data on various aspects including commute volume, land use patterns, pedestrian behaviours, and accident occurrences. Analysis of this data brought to light a host of critical problems such as inadequate urban planning, deficient road markings, and improper land utilization. Particularly during peak hours, pedestrian volumes surpassed manageable thresholds, exacerbating congestion.

In response, a solution emerged in the form of a proposed three-phase traffic signal design. This innovative approach aimed to alleviate congestion by strategically reducing conflict points, managing pedestrian-vehicle interactions, and minimizing delays. Implementation of this system held the promise of transforming the traffic dynamics of the road, potentially leading to a reduction in accident rates.

The application of the Webster's method facilitated the design process, resulting in a detailed signal diagram tailored to the intersection's unique requirements. Peak hours, identified as 07:00-11:00 in the morning, served as crucial time frames for data collection and analysis. Successful conversion of Passenger Car Unit (PCU) values revealed a peak value of 723, while an optimum cycle length of 87 seconds for all phases was determined.
Suggestions for improvement encompassed enhancing intersection planning, upgrading parking facilities, optimizing traffic flow management, and implementing effective signalization measures. This suggested approach has the power to set up a consistent set of steps that can make urban intersections better in developing countries, and ensure these improvements are long-lasting and don't harm the environment, paving the way for safer and more efficient transportation systems amid burgeoning urban populations.

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